

5. [Maximum mark: 7]

The first four terms of the Maclaurin series expansion of $(1 - x)^{-4}$ are

$$1 + ax + bx^2 + 20x^3, \ a, b \in \mathbb{Z}^+.$$

- (a) (i) Show that $a = 4$.

- (ii) Find the value of b .

[5]

A car was purchased four years ago. The car depreciated in value by 10% each year. The value of the car today is \$1000.

- (b) Using the results of part (a), estimate the value of the car four years ago.

[2]

[illegible]

Turn over